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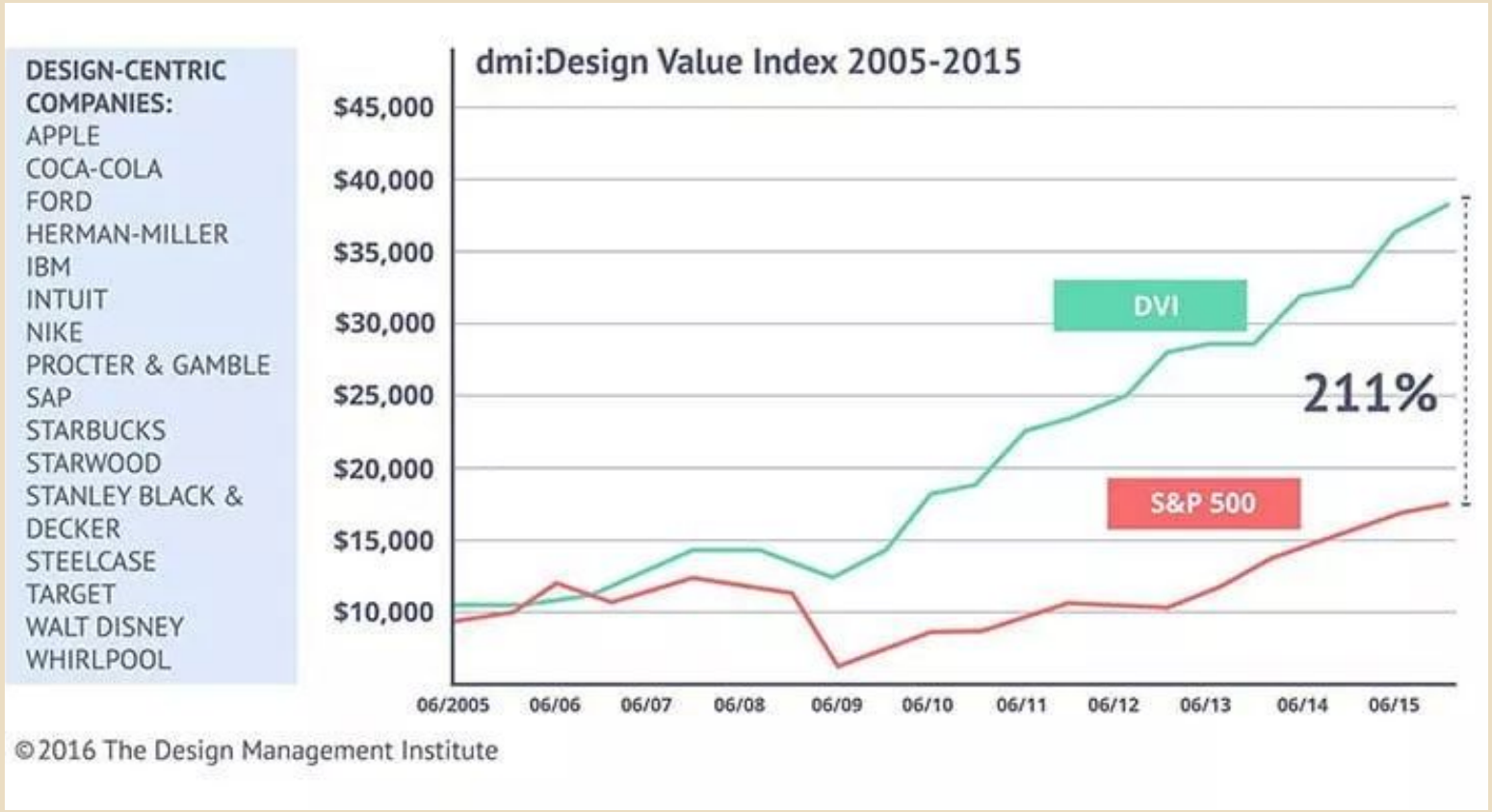
DESIGNING INNOVATIVE SOLUTIONS

INFORMATICS INSTITUTE OF TECHNOLOGY

LECTURE 2:

DESIGN THINKING AND PREPARING YOUR MIND FOR INNOVATION

DESIGN THINKING AS A STRATEGY FOR INNOVATION



Successful companies are always looking for new ways to compete in the marketplace. One approach, which has taken the corporate world by storm, is design thinking.

DESIGN THINKING AS A STRATEGY FOR INNOVATION

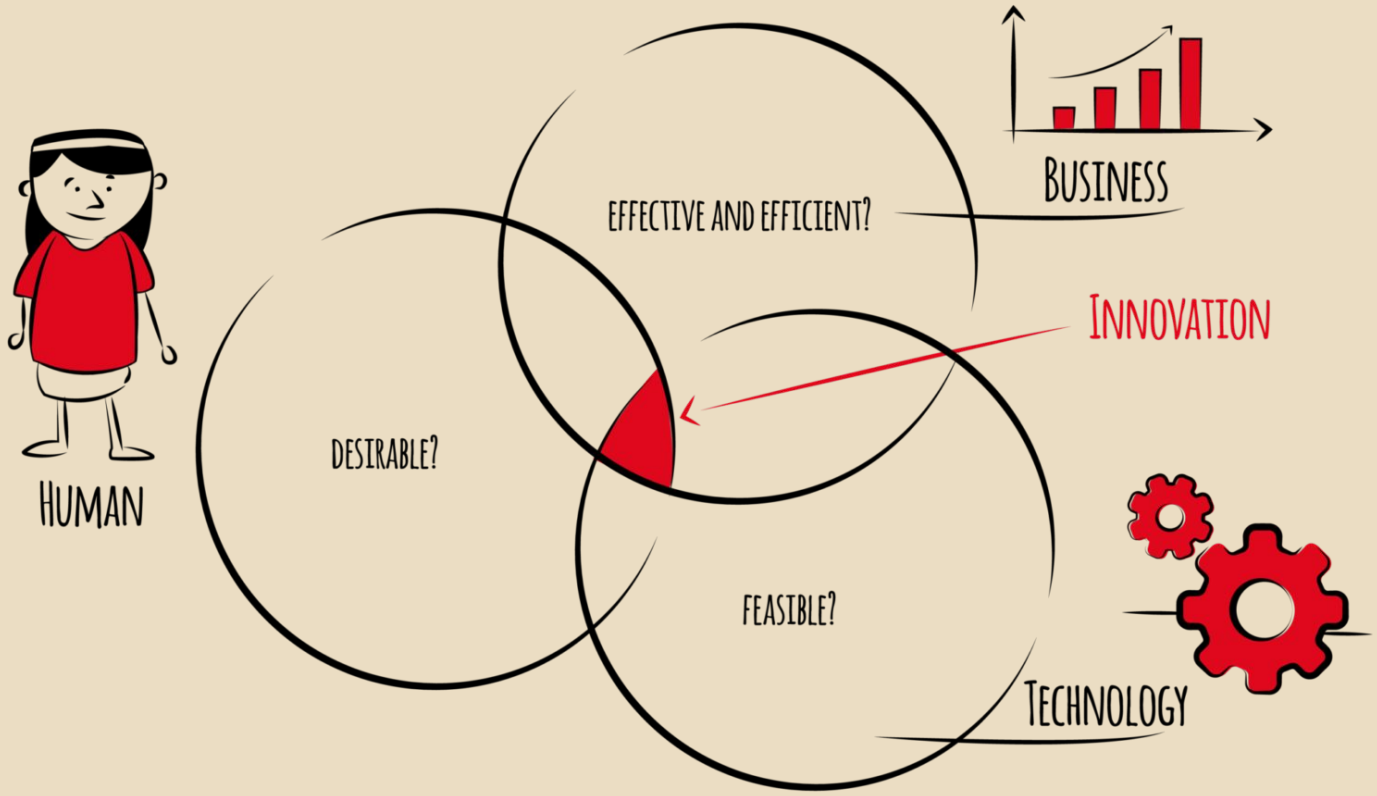
- When design principles are applied to strategy and innovation the success rate for innovation dramatically improves.
- Design-led companies such as Apple, Coca-Cola, IBM, Nike, Procter & Gamble and Whirlpool have outperformed the S&P 500 over the past 10 years by an extraordinary 211%, according to a 2015 assessment by the Design Management Institute.
- Due to the remarkable success rate of design-led companies, design has evolved beyond making objects.
- Organizations now want to learn how to think like designers and apply design principles to the workplace itself.
- Design thinking is at the core of effective strategy development and organizational change.

WHAT IS DESIGN THINKING?

- Design Thinking is a methodology used by designers to solve complex problems, and find desirable solutions for clients.
- A design mindset is not problem-focused, its solution focused and action oriented towards creating a preferred future.
- Design Thinking draws upon logic, imagination, intuition, and systemic reasoning, to explore possibilities of what could be—and to create desired outcomes that benefit the end user (the customer).

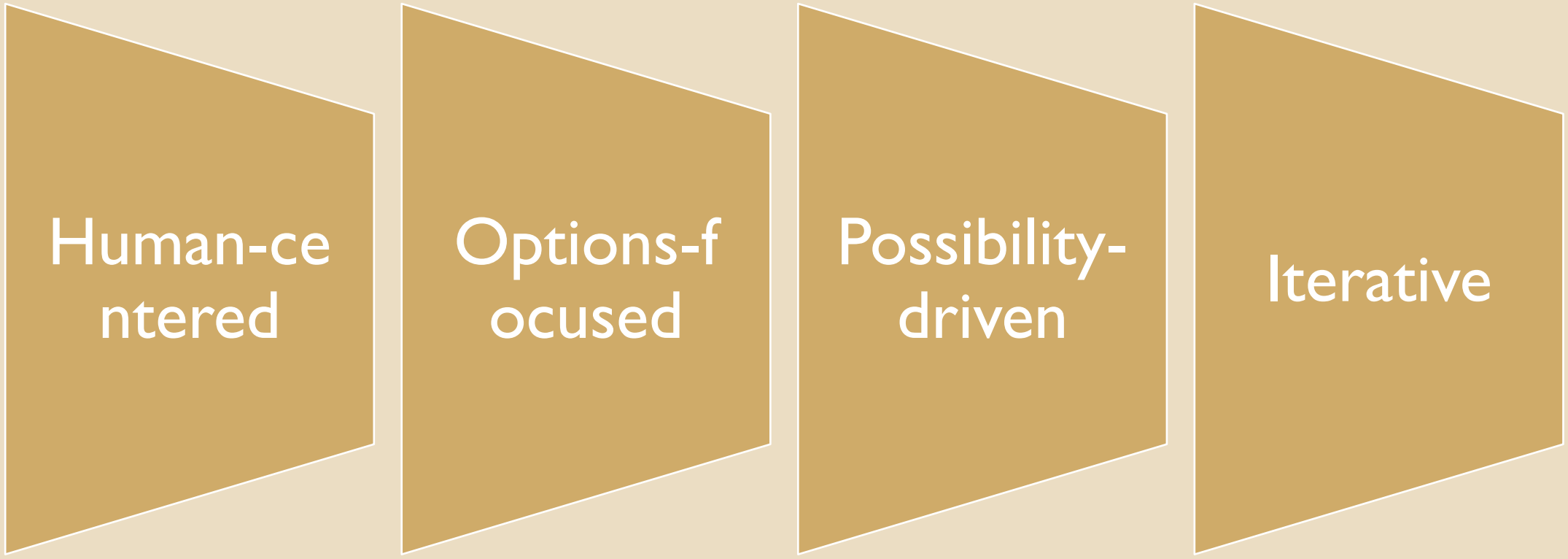
"Design thinking can be described as a discipline that uses the designer's sensibility and methods to match people's needs with what is technologically feasible and what a viable business strategy can convert into customer value and market opportunity." – Tim Brown CEO, IDEO

DESIGN THINKING



Great design has that "wow" factor that makes products more desirable and services more appealing to users. This explains why McKinsey, Deloitte, IBM, Facebook and Capital One have recently bought design firms.

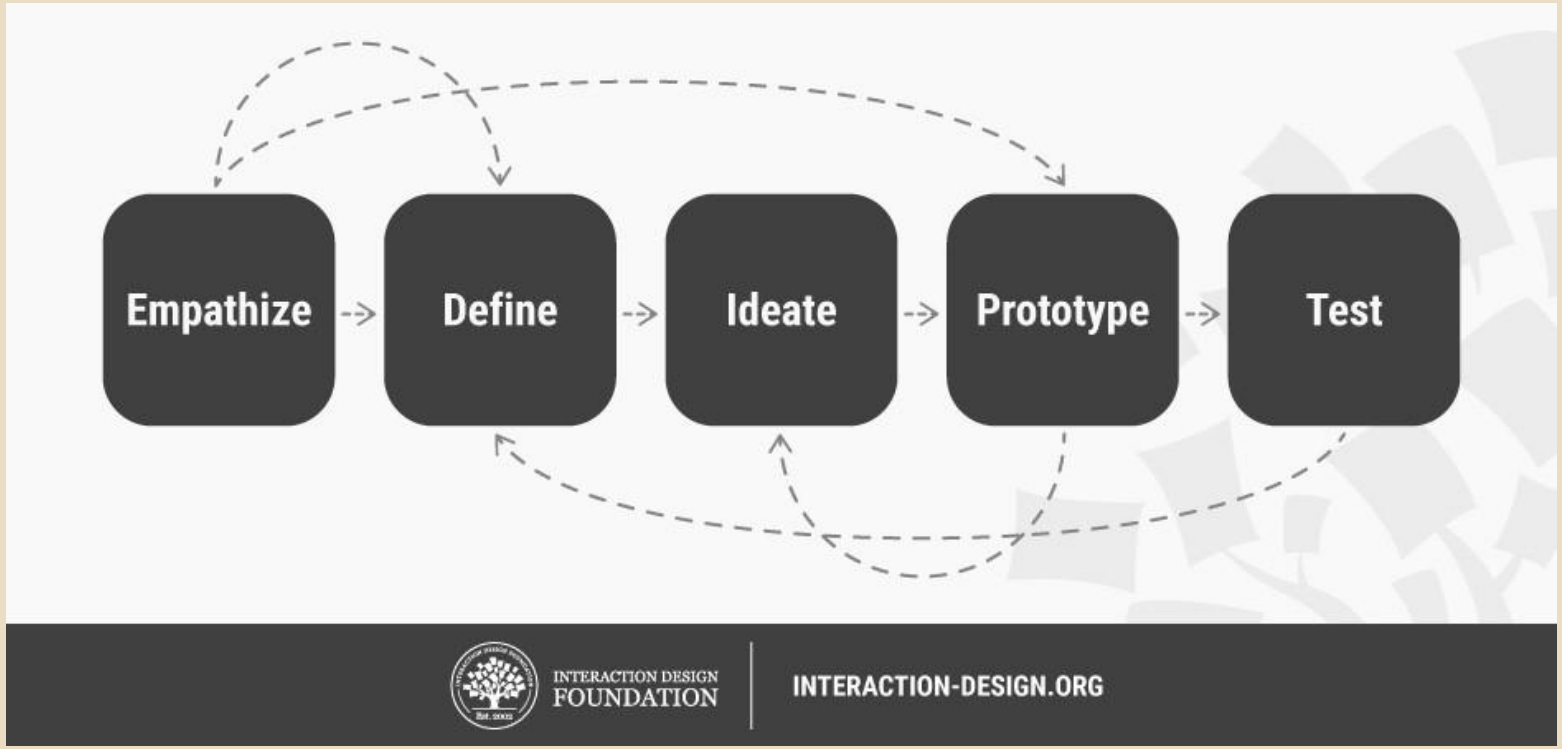
DESIGN THINKING



	Design thinking is appropriate if …	Linear analytic methods may be better if …
Is the problem human-centered?	Deep understanding of the actual people (users) involved	There are few human beings involved in the problem or the solution
How clearly do you understand the problem itself?	We need to explore and get agreement	We understand the problem clearly and are sure we're solving the right one
What's the level of uncertainty?	There are many unknowns (large and small), and past data is unlikely to help us	The past is a good predictor of the future
What data is already available to you?	There is very little relevant existing data to analyze	There are several clear sources of analogous data

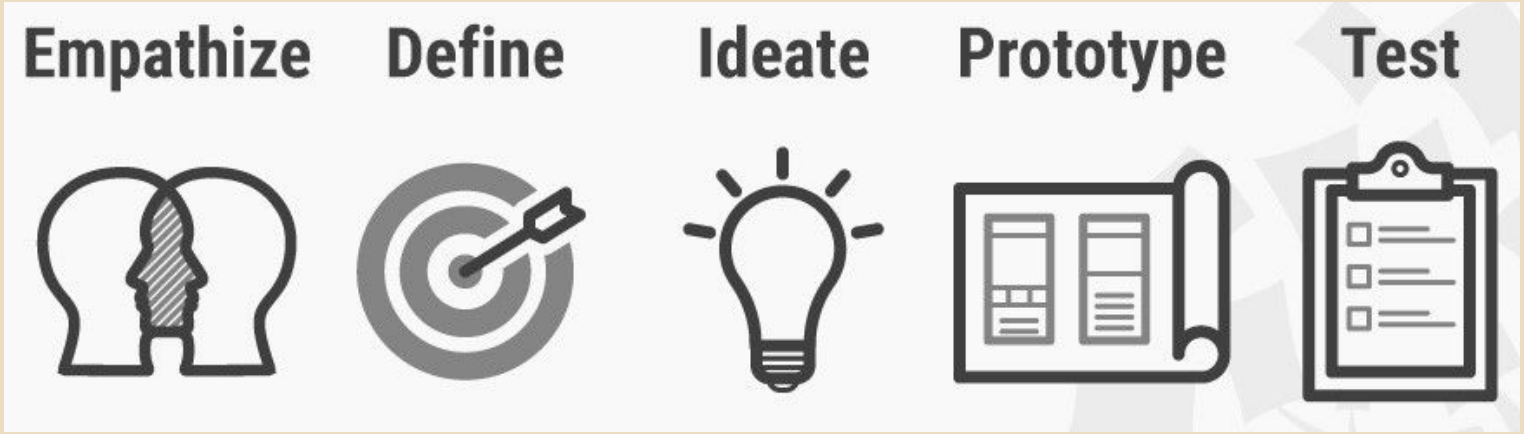
DESIGN THINKING – 5 STAGE PROCESS

Design thinking is a non-linear, iterative process which seeks to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.



DESIGN THINKING – 5 STAGE PROCESS

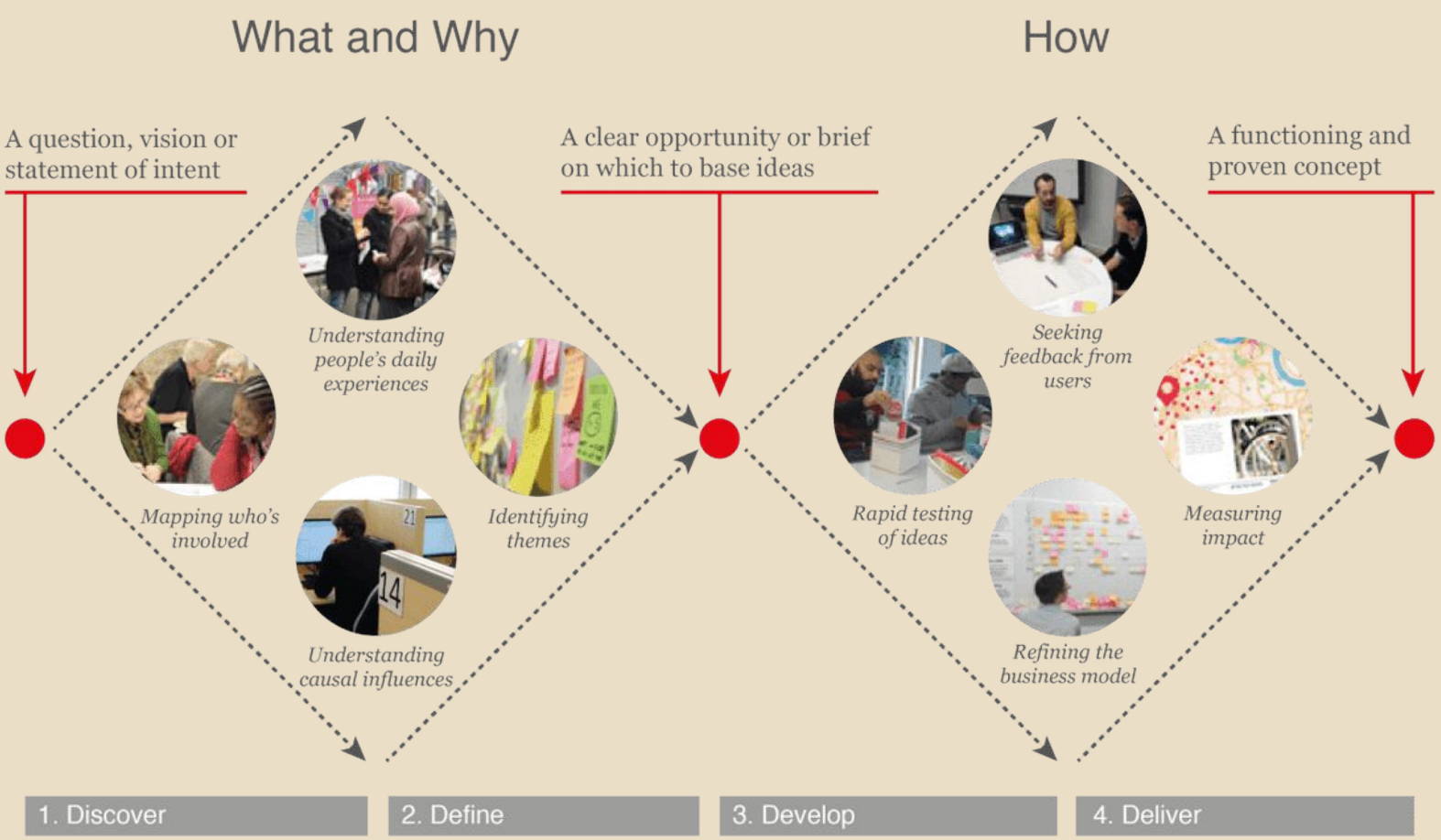
- Stage 1: Empathize—Research Your Users' Needs
- Stage 2: Define—State Your Users' Needs and Problems
- Stage 3: Ideate—Challenge Assumptions and Create Ideas
- Stage 4: Prototype—Start to Create Solutions
- Stage 5: Test—Try Your Solutions Out



A FRAMEWORK FOR DESIGN THINKING



CREATIVITY IS CENTRAL TO THE DESIGN PROCESS



This diagram illustrates the design thinking framework created by The Design Council (UK), which maps the design process into four distinct phases.

DESIGN THINKING INFORMS HUMAN-CENTERED INNOVATION

- Human-centered innovation begins with developing an understanding of customers' or users' unmet or unarticulated needs.
- The most secure source of new ideas that have true competitive advantage, and hence, higher margins, is customers' unarticulated needs.
- Customer intimacy—a deep knowledge of customers and their problems—helps to uncover those needs.
- Design thinking minimizes the uncertainty and risk of innovation by engaging customers or users through a series of prototypes to learn, test and refine concepts.
- Design thinkers rely on customer insights gained from real-world experiments, not just historical data or market research.

SELF-ANALYSIS

Select the most suitable answer which represents your mindset (Be honest to yourself!)

1. Challenges
   a) Avoid challenges
   b) Embrace challenges

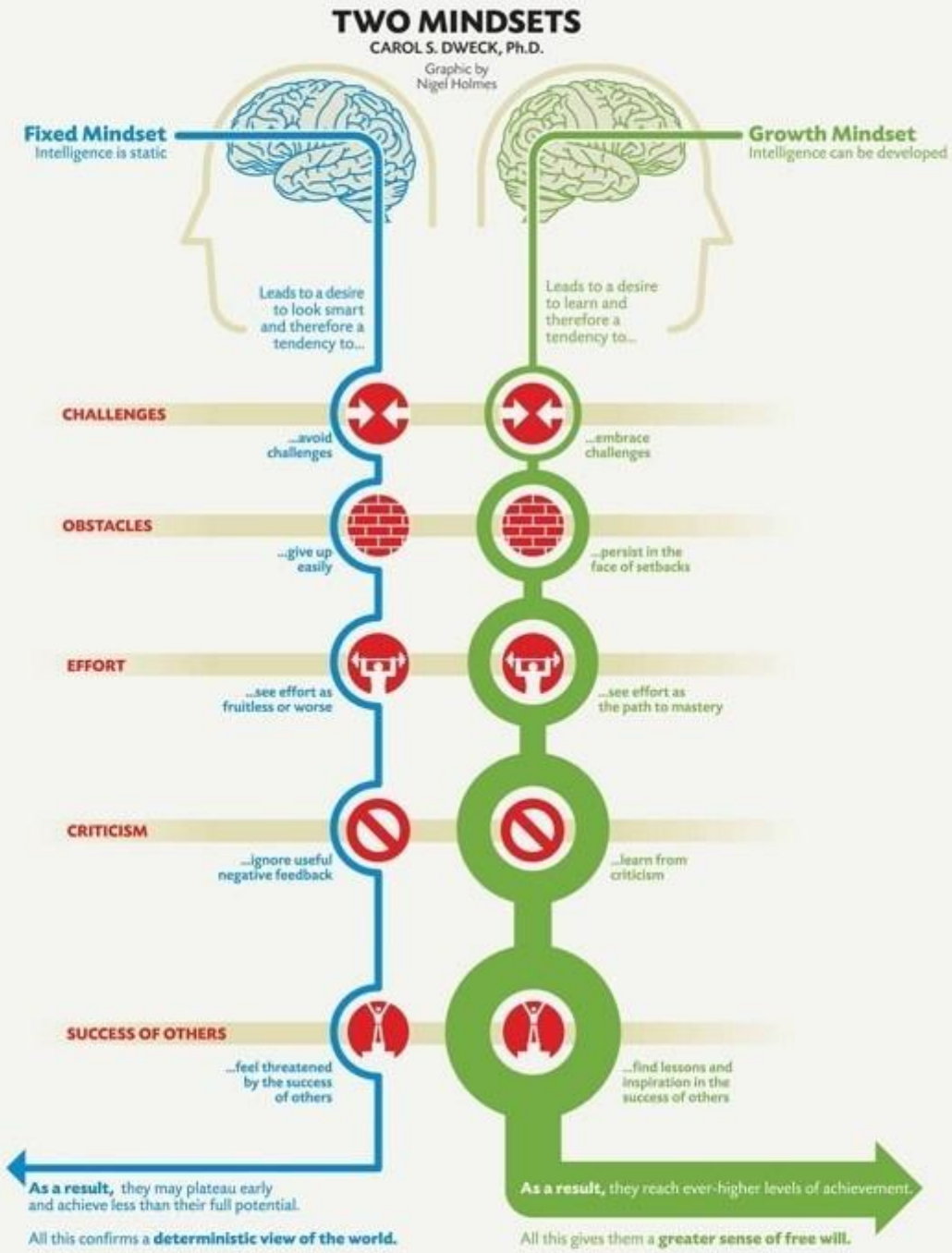
2. Obstacles
   a) Give up easily
   b) Persist in the face of set backs

3. Effort
   a) See effort as fruitless or worse
   b) See effort as the path to mastery

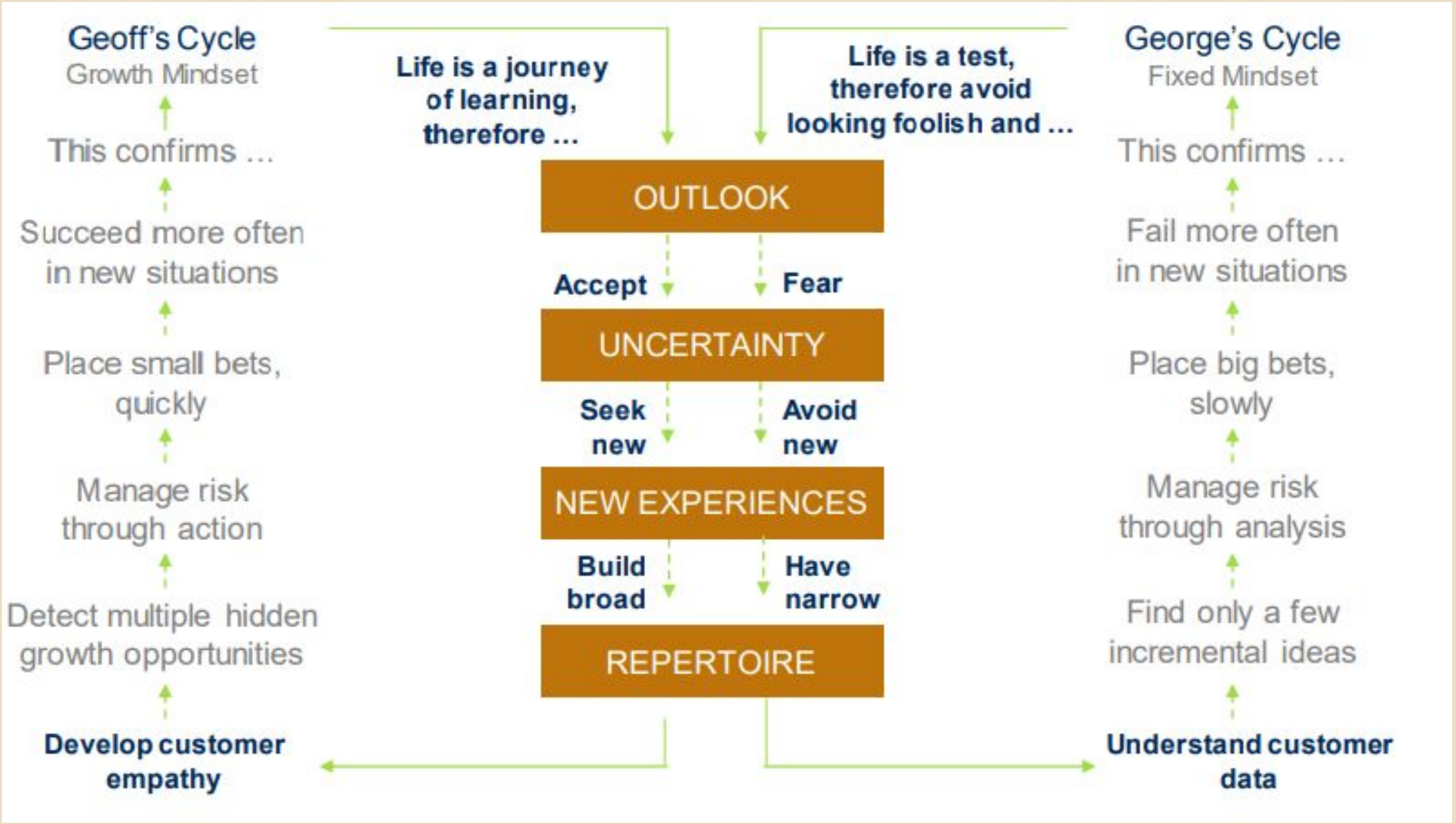
SELF-ANALYSIS

4. Criticism
   a) Ignore useful and negative feedback
   b) Learn from criticism

5. Success of others
   a) Feel threatened by the success of others
   b) Find lessons and inspiration in the success of others



Design thinking requires a growth mindset, because, in essence, it is a journey of learning and experimentation.


Geoff's Cycle
Growth Mindset
This confirms …
Succeed more often in new situations
Place small bets, quickly
Manage risk through action
Detect multiple hidden growth opportunities
Develop customer empathy
Life is a journey of learning, therefore …
Life is a test, therefore avoid looking foolish and …
OUTLOOK
Accept
Fear
UNCERTAINTY
Seek new
Avoid new
NEW EXPERIENCES
Build broad
Have narrow
REPERTOIRE
George's Cycle
Fixed Mindset
This confirms …
Fail more often in new situations
Place big bets, slowly
Manage risk through analysis
Find only a few incremental ideas
Understand customer data

A PREPARED MIND

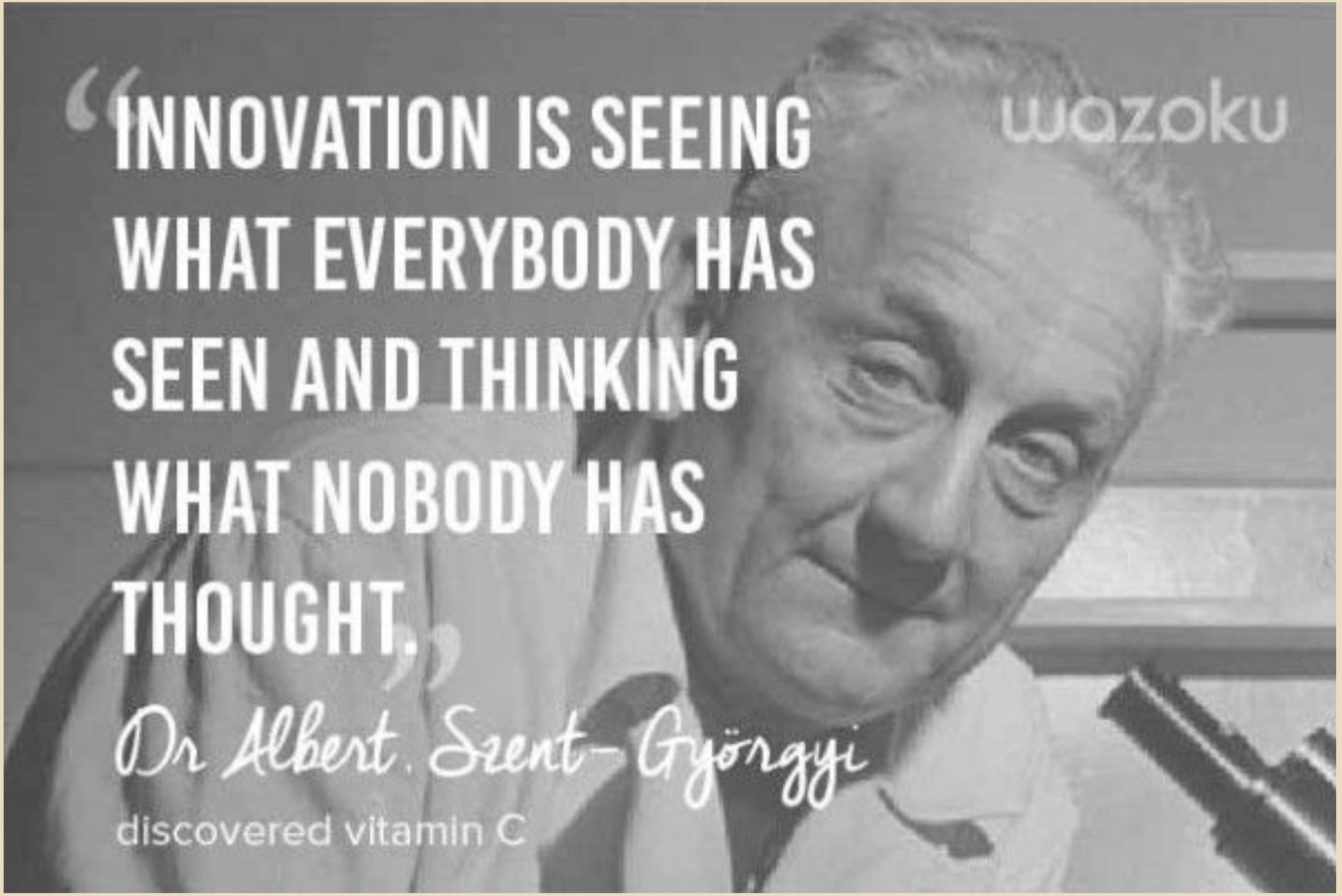
+ learning mindset

+ broad repertoire

+ customer empathy



= a prepared mind


"INNOVATION IS SEEING WHAT EVERYBODY HAS SEEN AND THINKING WHAT NOBODY HAS THOUGHT."
wazoku
Dr Albert Szent-Györgyi
discovered vitamin C